

Course: Large Language Models

Assignment 1

Deadline: 18 September,2025

Task 1: Binary classification using KNN classifier.

Dataset Description:

You are given a CSV file “data.csv” containing the measurements from a digitized image of a fine needle aspirate (FNA) of a breast mass. Divide the 80% data for training and 20% data for testing.

Independent variables/ Attributes/ Features:

1. Radius_mean
2. Texture_mean
3. Perimeter_mean
4. Area_mean
5. Smoothness_mean
6. Compactness_mean
7. Concavity_mean
8. Concavepoints_mean
9. Symmetry_mean
10. Fractal_dimension_mean
11. Radius_se
12. Texture_se
13. Perimeter_se
14. Area_se
15. Smoothness_se
16. Compactness_se
17. Concavity_se
18. Concavepoints_se
19. Symmetry_se
20. Fractal_dimension_se
21. Radius_worst
22. Texture_worst
23. Perimeter_worst
24. Area_worst
25. Smoothness_worst
26. Compactness_worst
27. Concavity_worst
28. concave points_worst

29. Symmetry_worst

30. Fractal_dimension_worst

Dependent variable/ Target Attribute/ Class:

i. "Diagnosis": (M = malignant, B = benign)

Model:

Build KNN based classification model from scratch to be trained on a given dataset and recommend the best **value for K** (number of neighbors) and **distance matrix** that produces the highest accuracy for test data.

Hyperparameters for Experimentation

- Values of K: 3,4,9,20 and 47.
- Distance Matrix: Euclidean Distance, Manhattan Distance, Minkowski Distance, Cosine Similarity, Hamming Distance .

Expected Outcomes:

- Pick the best value of K and distance matrix based on the testing accuracy, and for the best value of K, print confusion matrix, recall, and precision.
- Plot the graph between K values vs Accuracy considering different distance matrices.
- Inferences on the plots and inferences on the results observed.

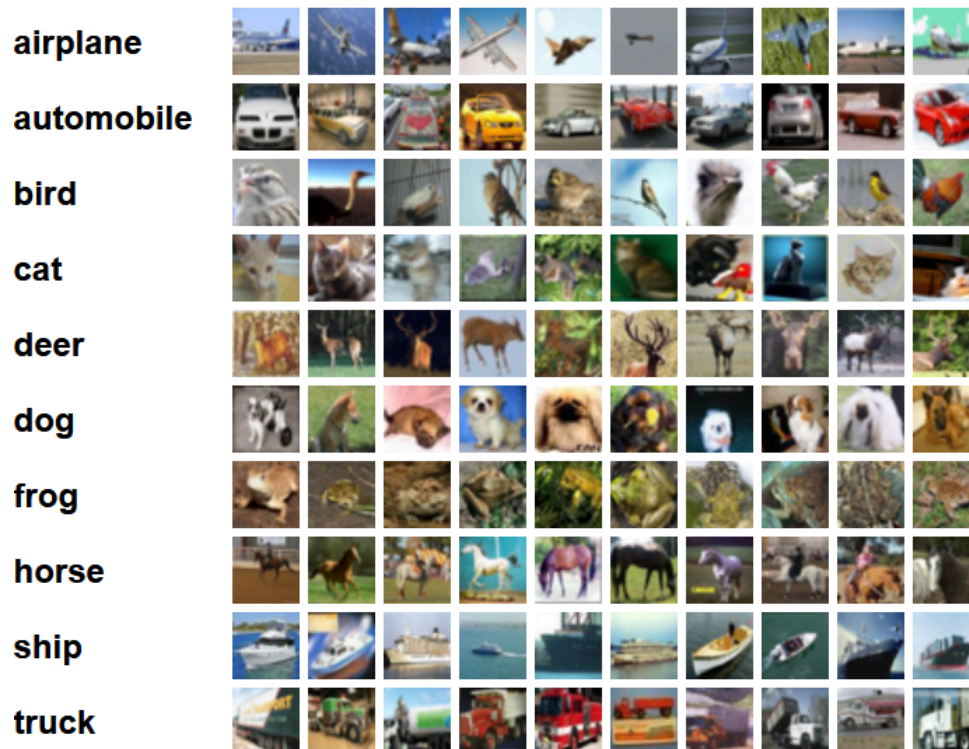
Bonus : Visualize the decision boundary for the best KNN model according to your results.

Task 2: Multi class classification using KNN classifier

CIFAR10 dataset

Dataset Description

The CIFAR-10 dataset is a collection of 60,000 32x32 color images in 10 classes, with 6,000 images per class. The dataset is split into 50,000 training images and 10,000 test images. Download [Link](#) .



Model:

Build KNN based classification model from scratch to classify these images into one of the 10 classes.

Expected Outcomes:

- Build a KNN-based classification model from scratch to classify images in the CIFAR-10 dataset.
- Experiment with different values of K (number of neighbors) and various distance metrics, including Euclidean Distance, Manhattan Distance, Minkowski Distance, Cosine Similarity, and Hamming Distance.
- Pick the best value of K and distance matrix based on the testing accuracy, and for the best value of K, print confusion matrix, recall, and precision.

- Visualize the impact of different distance metrics on the accuracy across varying values of K.
- Insights and inferences based on the experimental results and plots.

Note:

- You are not supposed to use pre-built functions for models from the libraries like sklearn, pytorch.
- Students will also be rewarded extra for solving bonus questions, thorough experimentation and insightful analysis based on the results and graphs they produce.

Report should be in PDF form and the report by a team should also include the observations about the results of studies.

Instruction:

Upload all your codes(.ipynb file) and reports(.pdf file) in a single zip file.

• **Give the name of the folder as Group_Assignment1 Example:**
Group01_Assignment1

• **Give the name of the zip file as Group_Assignment1.zip Example:**
Group01_Assignment1.zip

We will not accept the submission if you don't follow the above instructions